

# 1

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## Introduction

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### Chapter Objectives

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- Understanding the basic concepts of Total Quality Management (TQM)
- Brief overview of TQM framework
- Contribution of quality gurus in the TQM journey
- Benefits of TQM

### Definition

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Total Quality Management (TQM) is an enhancement to the traditional way of doing business. It is a proven technique to guarantee survival in world-class competition. Only by changing the actions of management will the culture and actions of an entire organization be transformed. TQM is for the most part common sense. Analyzing the three words, we have:

*Total*—Made up of the whole.

*Quality*—Degree of excellence a product or service provides.

*Management*—Act, art, or manner of handling, controlling, directing, etc.

Therefore, TQM is the art of managing the whole to achieve excellence. The Golden Rule is a simple, but effective way to explain it: Do unto others as you would have them do unto you. /

TQM is defined as both a philosophy and a set of guiding principles that represent the foundation of a continuously improving organization. It is the application of quantitative methods and human resources to improve all the processes within an organization and exceed customer needs now and in the future. TQM integrates fundamental management techniques, existing improvement efforts, and technical tools under a disciplined approach.

TABLE 1-1  
**New and Old Cultures**

| <i>Quality Element</i> | <i>Previous State</i>              | <i>TQM</i>                              |
|------------------------|------------------------------------|-----------------------------------------|
| Definition             | Product-oriented                   | Customer-oriented                       |
| Priorities             | Second to service and cost         | First among equals of service and cost  |
| Decisions              | Short-term                         | Long-term                               |
| Emphasis               | Detection                          | Prevention                              |
| Errors                 | Operations                         | System                                  |
| Responsibility         | Quality control                    | Everyone                                |
| Problem Solving        | Managers                           | Teams                                   |
| Procurement            | Price                              | Life-cycle costs, partnership           |
| Manager's Role         | Plan, assign, control, and enforce | Delegate, coach, facilitate, and mentor |

6. Performance measures such as uptime, percent nonconforming, absenteeism, and customer satisfaction should be determined for each functional area. These measures should be posted for everyone to see. Quantitative data are necessary to measure the continuous quality improvement activity.

The purpose of TQM is to provide a quality product and/or service to customers, which will, in turn, increase productivity and lower cost. With a higher quality product and lower price, competitive position in the marketplace will be enhanced. This series of events will allow the organization to achieve the objectives of profit and growth with greater ease. In addition, the work force will have job security, which will create a satisfying place to work.

As previously stated, TQM requires a cultural change. Table 1-1 compares the previous state with the TQM state for typical quality elements. This change is substantial and will not be accomplished in a short period of time. Small organizations will be able to make the transformation much faster than large organizations.

## Gurus of Total Quality Management

### Shewhart

Walter A. Shewhart, PhD, spent his professional career at Western Electric and Bell Telephone Laboratories, both divisions of AT&T. He developed control chart theory with control limits, assignable and chance causes of variation, and rational subgroups (see Chapter 18). In 1931, he authored *Economic Control of Quality of Manufactured Product*, which is regarded as a complete and thorough work of the basic principles of quality control. He also developed the PDCA cycle for learning and improvement (see Chapter 6).

## Crosby

Phillip B. Crosby authored his first book, *Quality Is Free*, in 1979, which was translated into 15 languages. It sold 1.5 million copies and changed the way management looked at quality. He argued that "doing it right the first time" is less expensive than the costs of detecting and correcting nonconformities. In 1984, he authored *Quality Without Tears*, which contained his four absolutes of quality management. These absolutes are: quality is conformance to requirements, prevention of nonconformance is the objective not appraisal, the performance standard is zero defects not "that's close enough," and the measurement of quality is the cost of nonconformance.

## Taguchi

Genichi Taguchi, PhD, developed his loss function concept that combines cost, target, and variation into one metric. Because the loss function is reactive, he developed the signal to noise ratio as a proactive equivalent. The cornerstone of Taguchi's philosophy is the robust design of parameters and tolerances. It is built on the simplification and use of traditional design of experiments. These concepts are described in Chapter 20.

## TQM Framework

Figure 1-1 shows the framework for the TQM system. It begins with the knowledge provided by gurus of quality: Shewhart, Deming, Juran, Feigenbaum, Ishikawa, Crosby, and Taguchi. As the figure shows, they contributed to the development of principles and practices and/or the tools and techniques.

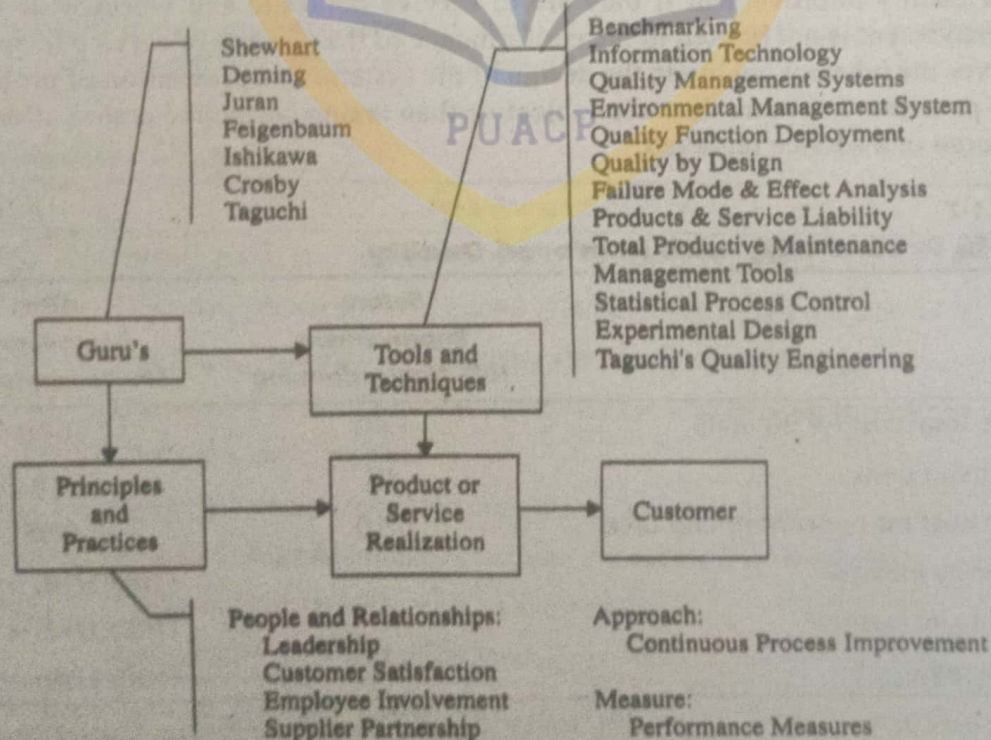


Figure 1-1 TQM Framework

TQM does not occur overnight; there are no quick remedies. It takes a long time to build the appropriate emphasis and techniques into the culture. Overemphasis on short-term results and profits must be set aside so long-term planning and constancy of purpose will prevail.

## Defining Quality

When the expression "quality" is used, we usually think in terms of an excellent product or service that fulfills or exceeds our expectations. These expectations are based on the intended use and the selling price. For example, a customer expects a different performance from a plain steel washer than from a chrome-plated steel washer because they are a different grade. When a product surpasses our expectations we consider that quality. Thus, it is somewhat of an intangible based on perception. Quality can be quantified as follows:

$$Q = P/E$$

where  $Q$  = quality  
 $P$  = performance  
 $E$  = expectations

If  $Q$  is greater than 1.0, then the customer has a good feeling about the product or service. Of course, the determination of  $P$  and  $E$  will most likely be based on perception with the organization determining performance and the customer determining expectations.

A more definitive definition of quality is given in ISO 9000: 2000. It is defined as the degree to which a set of inherent characteristics fulfills requirements. *Degree* means that quality can be used with adjectives such as poor, good, and excellent. *Inherent* is defined as existing in something, especially as a permanent characteristic. *Characteristics* can be quantitative or qualitative. *Requirement* is a need or expectation that is stated; generally implied by the organization, its customers, and other interested parties; or obligatory.

Quality has nine different dimensions. Table 1-3 shows these nine dimensions of quality with their meanings and explanations in terms of a overhead projector.

TABLE 1-3  
**The Dimensions of Quality**

| Dimension   | Meaning and Example                                                     |
|-------------|-------------------------------------------------------------------------|
| Performance | Primary product characteristics, such as the brightness of the picture  |
| Features    | Secondary characteristics, added features, such as remote control       |
| Conformance | Meeting specifications or industry standards, workmanship               |
| Reliability | Consistency of performance over time, average time for the unit to fail |
| Durability  | Useful life, includes repair                                            |
| Service     | Resolution of problems and complaints, ease of repair                   |
| Response    | Human-to-human interface, such as the courtesy of the dealer            |
| Aesthetics  | Sensory characteristics, such as exterior finish                        |
| Reputation  | Past performance and other intangibles, such as being ranked first      |

Adapted from David A. Garvin, *Managing Quality: The Strategic and Competitive Edge* (New York: Free Press, 1988).

## Obstacles

Implementation of TQM is described in the next chapter, on leadership. This section gives information concerning the obstacles associated with implementation.

Many organizations, especially small ones with a niche, are comfortable with their current state. They are satisfied with the amount of work being performed, the profits realized, and the perception that the customers are satisfied. Organizations with this culture will see little need for TQM until they begin to lose market share.

Once an organization embarks on TQM, there will be obstacles to its successful implementation. The first eight most common were determined by Robert J. Masters after an extensive literature search and the last obstacle added by the authors.<sup>6</sup> They are given below.

### Lack of Management Commitment

In order for any organizational effort to succeed, there must be a substantial management commitment of management time and organizational resources. The purpose must be clearly and continuously communicated to all personnel. Management must consistently apply the principles of TQM.

Robert Galvin of Motorola said that only the CEO can ensure, even in times of great pressure, that quality and customer satisfaction are preserved. In a survey of 188 quality professionals, 66% reported that management's compensation is not linked to quality goals such as failure costs, customer complaints, and cycle time reduction.<sup>7</sup>

### Inability to Change Organizational Culture

Changing an organization's culture is difficult and will require as much as five years. Individuals resist change—they become accustomed to doing a particular process and it becomes the preferred way. Management must understand and utilize the basic concepts of change. They are:

1. People change when they want to and to meet their own needs.
2. Never expect anyone to engage in behavior that serves the organization's values unless adequate reason (why) has been given.
3. For change to be accepted, people must be moved from a state of fear to trust.

It is difficult for individuals to change their way of doing things; it is much more difficult for an organization to make a cultural change.

Management by exhortation and inspiration will fail. Speeches, slogans, and campaigns that are supposed to motivate people are only effective for a short period of time. Impediments to a cultural change are the lack of effective communication and emphasis on short-term results. Organizations that spend more time planning for the cultural aspects of implementing a TQM program will improve their chances of success.<sup>8</sup>

<sup>6</sup> Robert J. Masters, "Overcoming the Barriers to TQM's Success," *Quality Progress* (May 1996): 53–55.

<sup>7</sup> Nabil Tamimi and Rose Sebastianelli, "The Barrier to Total Quality Management," *Quality Progress* (June 1998): 57–60.

<sup>8</sup> Gary Salegna and Farzaneh Fasel, "Obstacles to Implementing Quality," *Quality Progress* (July 2000): 53–57.

One way to overcome this obstacle is to give the right people direct access to the customers. Maruti Suzuki, the leading car manufacturing company in India, takes significant efforts in training their service technicians and dealers' sales staff in order to ensure that their actions and interactions are in synchronization with the changes in customer profiles and expectations.<sup>10</sup> When an organization fails to empower individuals and teams, it cannot hold them responsible for producing results.

### Inadequate Use of Empowerment and Teamwork

Teams need to have the proper training and, at least in the beginning, a facilitator. Whenever possible, the team's recommendations should be followed. Individuals should be empowered to make decisions that affect the efficiency of their process or the satisfaction of their customers. Solar Turbine Inc. flattened its organization by restructuring into work teams and delegating authority to the point of customer contact or to the work performed.

### Failure to Continually Improve

It is tempting to sit back and rest on your laurels. However, a lack of continuous improvement of the processes, product, and/or service will even leave the leader of the pack in the dust. Will Rogers said it best, "Even if you're on the right track, you'll get run over if you just sit there." Even though Champion Mortgage's 1998 business volume increased 59%, it continues to address culture, staff, and service issues.<sup>11</sup>

### Benefits of TQM

According to a survey of manufacturing firms in Georgia, the benefits of TQM are improved quality, employee participation, teamwork, working relationships, customer satisfaction, employee satisfaction, productivity, communication, profitability, and market share.<sup>12</sup>

TQM is a good investment as shown by a ten-year study by Hendricks and Singhal. They showed that there is a strong link between TQM and financial performance. The researchers selected a group of 600 publicly traded organizations that had won awards for effectively implementing TQM. They then selected a control group similar in size and industry to the award winners. Performance of both groups was compared during the five years prior to the award and five years after winning the award. No difference was shown between the two groups prior to the award. However, as shown below the award group far outstripped the control group during the five-year period after the award.

| Description                | Control | Award |
|----------------------------|---------|-------|
| Growth in Operating Income | 43%     | 91%   |
| Increase in Sales          | 32%     | 69%   |
| Increase in Total Assets   | 37%     | 79%   |

<sup>10</sup> Annual Report of Maruti Suzuki 2008-09.

<sup>11</sup> Mark R. Hagan, "Complacency—the Enemy of Quality," *Quality Progress* (October 1999): 37-44.

<sup>12</sup> Christopher M. Lowery, et. al., "TQM's Human Resource Component," *Quality Progress* (February 2000): 55-58.

The second step is *prevention*, which is the development of a system that will minimize the costs. Because management has a good idea of the appraisal costs, this step can proceed concurrently with Step 1.

- Pressure can be addressed by being involved in the development of goals and values and developing policies that allow for individual diversity, dissent, and decision-making input.
- Opportunity can be addressed by developing policies that encourage and protect whistleblowers and require the existence of ombudsmen who can work confidentially with people to solve ethical problems internally.
- Attitude can be addressed by requiring ethics training for all personnel, recognizing ethical conduct in the workplace, requiring performance appraisals to include ethics, and encouraging open discussion concerning ethical behavior issues.

The third step is *promotion*, which is the continuous advertising of ethical behavior in order to develop an ethical organizational culture that is clear, positive, and effective.

- To be clear the philosophy needs to be written, with input from all personnel, and posted. Standardized ethics training should be given to everyone to: (1) teach them how to clarify ethical issues; (2) encourage them to get the facts before acting; (3) encourage them to consider all the consequences before acting; and (4) show them how to test their actions in advance. This testing can be accomplished by asking: (1) Is it legal? (2) Is it right? (3) Is it beneficial for all involved? and (4) How would I feel if it was published on the front page of the newspaper?
- To be positive, the culture should be about doing what is right, encouraging principled organizational dissent, and rewarding ethical behavior.
- To be effective, the philosophy must be set and adopted by senior management, with input from all personnel. Senior management should act as they would want others to act and make no exceptions.

### Final Comment

Quality is dependent on ethical behavior. Doing what is right in the first place is a proven way to reduce costs, improve competitiveness, and create customer satisfaction. Many companies are hiring ethics consultants to help them achieve their goals.

Unethical behavior by Enron and Worldcom executives in 2002 resulted in bankruptcy for those companies.

In 2012, Kingfisher Airlines faced financial problems and had deferred the salary payment of its employees and taxes. It also started routinely cancelling scheduled flights resulting in serious loss of credibility.

### The Deming Philosophy<sup>a</sup> 14-steps

Deming's philosophy is given in his 14 points. Most of these points were given in a seminar for 21 Presidents of leading Japanese industry in 1950. The rest were developed and the original ones modified over a period of three decades.

<sup>a</sup> Adapted from *Out of the Crisis* by W. Edwards Deming by permission of MIT and the W. Edwards Deming Institute. Published by MIT, Center for Advanced Engineering Study, Cambridge, MA 02139. Copyright 1986 by W. Edwards Deming.

## 1. Create and Publish the Aims and Purposes of the Organization

Management must demonstrate constantly their commitment to this statement. It must include investors, customers, suppliers, employees, the community, and a quality philosophy. The statement is a forever-changing document that requires input from everyone. Organizations must develop a long-term view of at least ten years and plan to stay in business by setting long-range goals. Resources must be allocated for research, training, and continuing education to achieve the goals. Innovation is promoted to ensure that the product or service does not become obsolete. A family organizational philosophy is developed to send the message that everyone is part of the organization.

## 2. Learn the New Philosophy

Top management and everyone must learn the new philosophy. Organizations must seek never-ending improvement and refuse to accept nonconformance. Customer satisfaction is the number one priority, because dissatisfied customers will not continue to purchase nonconforming products and services. The organization must concentrate on defect prevention rather than defect detection. By improving the process, the quality and productivity will improve. Everyone in the organization, including the union, must be involved in the quality journey and change his or her attitude about quality. The supplier must be helped to improve quality by requiring statistical evidence of conformance and shared information relative to customer expectations.

## 3. Understand the Purpose of Inspection

Management must understand that the purpose of inspection is to improve the process and reduce its cost. For the most part, mass inspection is costly and unreliable. Where appropriate, it should be replaced by never-ending improvement using statistical techniques. Statistical evidence is required of self and supplier. Every effort should be made to reduce and then eliminate acceptance sampling. Mass inspection is managing for failure and defect prevention is managing for success.

## 4. Stop Awarding Business Based on Price Alone

The organization must stop awarding business based on the low bid, because price has no meaning without quality. The goal is to have single suppliers for each item to develop a long-term relationship of loyalty and trust, thereby providing improved products and services. Purchasing agents must be trained in statistical process control and require it from suppliers. They must follow the materials throughout the entire life cycle in order to examine how customer expectations are affected and provide feedback to the supplier regarding the quality.

## 5. Improve Constantly and Forever the System

Management must take more responsibility for problems by actively finding and correcting problems so that quality and productivity are continually and permanently improved and costs are reduced. The focus is on preventing problems before they happen. Variation is expected, but there must be a continual striving for its reduction using control charts. Responsibilities are assigned to teams to remove the causes of problems and continually improve the process.

## 6. Institute Training

Each employee must be oriented to the organization's philosophy of commitment to never-ending improvements. Management must allocate resources to train employees to perform their jobs in the best manner possible. Everyone should be trained in statistical methods, and these methods should be used to monitor the need for further training.

## 7. Teach and Institute Leadership

Improving supervision is management's responsibility. They must provide supervisors with training in statistical methods and these 14 points so the new philosophy can be implemented. Instead of focusing on a negative, fault-finding atmosphere, supervisors should create a positive, supportive one where pride in workmanship can flourish. All communication must be clear from top management to supervisors to operators.

## 8. Drive Out Fear, Create Trust, and Create a Climate for Innovation

Management must encourage open, effective communication and teamwork. Fear is caused by a general feeling of being powerless to control important aspects of one's life. It is caused by a lack of job security, possible physical harm, performance appraisals, ignorance of organization goals, poor supervision, and not knowing the job. Driving fear out of the workplace involves managing for success. Management can begin by providing workers with adequate training, good supervision, and proper tools to do the job, as well as removing physical dangers. When people are treated with dignity, fear can be eliminated and people will work for the general good of the organization. In this climate, they will provide ideas for improvement.

## 9. Optimize the Efforts of Teams, Groups, and Staff Areas

Management must optimize the efforts of teams, work groups, and staff areas to achieve the aims and purposes of the organization. Barriers exist internally among levels of management, among departments, within departments, and among shifts. Externally, they exist between the organization and its customers and suppliers. These barriers exist because of poor communication, ignorance of the organization's mission, competition, fear, and personal grudges or jealousies. To break down the barriers, management will need a long-term perspective. All the different areas must work together. Attitudes need to be changed; communication channels opened; project teams organized; and training in teamwork implemented. Multifunctional teams, such as used in concurrent engineering, are an excellent method.

## 10. Eliminate Exhortations for the Work Force

Exhortations that ask for increased productivity without providing specific improvement methods can handicap an organization. They do nothing but express management's desires. They do not produce a better product or service, because the workers are limited by the system. Goals should be set that are achievable and are committed to the long-term success of the organization. Improvements in the process cannot be made unless the tools and methods are available.

### 11a. Eliminate Numerical Quotas for the Work Force

Instead of quotas, management must learn and institute methods for improvement. Quotas and work standards focus on quantity rather than quality. They encourage poor workmanship in order to meet their quotas. Quotas should be replaced with statistical methods of process control. Management must provide and implement a strategy for never-ending improvements and work with the work force to reflect the new policies.

### 11b. Eliminate Management by Objective

Instead of management by objective, management must learn the capabilities of the processes and how to improve them. Internal goals set by management, without a method, are a burlesque. Management by numerical goal is an attempt to manage without knowledge of what to do. An excellent analysis supporting this point is given by Castellano and Roehm.<sup>9</sup>

### 12. Remove Barriers That Rob People of Pride of Workmanship

Loss of pride in workmanship exists throughout organizations because: (1) workers do not know how to relate to the organization's mission, (2) they are being blamed for system problems, (3) poor designs lead to the production of 'junk,' (4) inadequate training is provided, (5) punitive supervision exists, and (6) inadequate or ineffective equipment is provided for performing the required work. Restoring pride will require a long-term commitment by management. When workers are proud of their work, they will grow to the fullest extent of their job. Management must give employees operational job descriptions, provide the proper tools and materials, and stress the workers' understanding of their role in the total process. By restoring pride, everyone in the organization will be working for the common good. A barrier for people on salary is the annual rating of performance.

### 13. Encourage Education and Self-Improvement for Everyone

What an organization needs is people who are improving with education. A long-term commitment to continuously train and educate people must be made by management. Deming's 14 points and the organization's mission should be the foundation of the education program. Everyone should be retrained as the organization requirements change to meet the changing environment.

### 14. Take Action to Accomplish the Transformation

Management has to accept the primary responsibility for the never-ending improvement of the process. It has to create a corporate structure to implement the philosophy. A cultural change is required from the previous "business as usual" attitude. Management must be committed, involved, and accessible if the organization is to succeed in implementing the new philosophy. Hillerich & Bradsby Co., the makers of the Louisville Slugger baseball bat, have used Deming's 14 points since 1985 and now have 70% of the professional baseball bat market.<sup>10</sup>

<sup>9</sup> Joseph F. Castellano and Harper A. Roehm, "The Problems With Managing By Objectives and Results," *Quality Progress* (March 2001): 39-46.

<sup>10</sup> March L. Jacques, "Big League Quality," *Quality Progress* (August 2001): 27-34.

# 3

## Customer Satisfaction

### Chapter Objectives

- Understanding what is customer satisfaction
- Identifying the customer: internal and external
- Factors which influence customer satisfaction
- Measurement of customer satisfaction and dissatisfaction
- Service quality and customer satisfaction
- Role of leader and frontline people
- Understanding the Kano model

### Introduction

The most important asset of any organization is its customers. An organization's success depends on how many customers it has, how much they buy, and how often they buy. Customers that are satisfied will increase in number, buy more, and buy more frequently. Satisfied customers also pay their bills promptly, which greatly improves cash flow—the lifeblood of any organization. The organizational diagram in Figure 3-1 best exemplifies just how important the customer is to any organization.

Increasingly, manufacturing and service organizations are using customer satisfaction as the measure of quality. The importance of customer satisfaction is not only due to national competition but also due to worldwide competition. This fact is reflected in Malcolm Baldrige National Quality Award where customer satisfaction has high importance in the criteria. Similarly, customer satisfaction standards are woven throughout ISO 9000: 2005. Customer satisfaction is one of the major purposes of a quality management system.<sup>1</sup>

Total Quality Management (TQM) implies an organizational obsession with meeting or exceeding customer expectations, so that customers are delighted. Understanding the customer's needs and expectations is essential to winning new business and keeping existing business. An organization

<sup>1</sup> Craig Cochran, "Customer Satisfaction: The Elusive Quarry," *Quality Digest* (November 2001): 45–50.

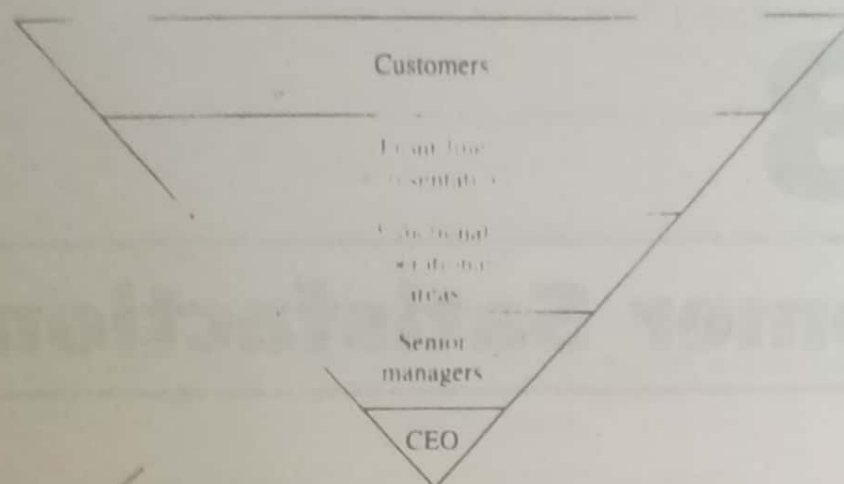


Figure 3-1 Customer Satisfaction Organizational Diagram

must give its customers a quality product or service that meets their needs at a reasonable price, which includes on-time delivery and outstanding service. To attain this level, the organization needs to continually examine their quality system to see if it is responsive to ever-changing customer requirements and expectations.

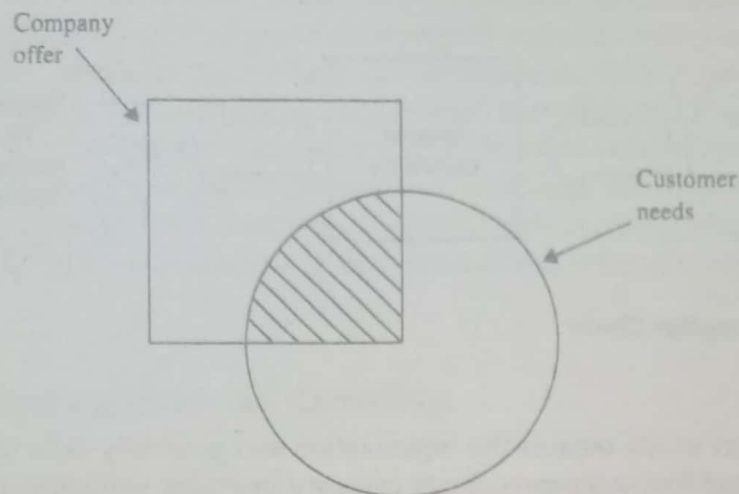
The most successful TQM programs begin by defining quality from the customer's perspective. As defined in Chapter 1, quality means meeting or exceeding the customer's expectations. Dr. Deming added that quality also means anticipating the future needs of the customer. Customer satisfaction, not increasing profits, must be the primary goal of the organization. It is the most important consideration, because satisfied customers will lead to increased profits.

A simplistic definition of customer satisfaction is illustrated by the Teboul model, which is shown in Figure 3-2. The customer's needs are represented by the circle, and the square depicts the product or service offered by the organization. Total satisfaction is achieved when the offer matches the need, or the circle is superimposed on the square. The goal is to cover the expected performance level better than the competitors.

That part of the square that lies within the circle is perceived by the customer as satisfying, and the part of the square outside the circle is perceived as unnecessary. It is important that the organization listen to the "voice of the customer" and ensure that its marketing, design, production, and distribution processes truly meet the expectations of the customer.

Customer satisfaction seems simple enough, and yet it is far from simple. Customer satisfaction is not an objective statistic but more of a feeling or attitude. Although certain statistical patterns can be developed to represent customer satisfaction, it is best to remember that people's opinions and attitudes are subjective by nature.

Because customer satisfaction is subjective, it is hard to measure. There are so many facets to a customer's experience with a product or service that need to be measured individually to get an accurate total picture of customer satisfaction. Whether or not a customer is satisfied cannot be classed as a yes or no answer. Errors can occur when customer satisfaction is simplified too much. The Teboul model, for instance, describes customer satisfaction as the degree to which the customer's experience of a service or product matches her expectations. Using this model, a customer's satisfaction level would be the same if the experience were mediocre in the context of low expectations, or if the experience were superior in the context of high expectations. Customer satisfaction's focus is creating superior experiences, not mediocre experiences.



**Figure 3-2 Customer Satisfaction Model**

Reproduced, with permission, from James Teboul, *Managing Quality Dynamics* (Englewood Cliffs, N.J.: Prentice Hall, Hemel Hempstead, 1991).

Since customer satisfaction is hard to measure, the measurement often is not precise. As with most attitudes, there is variability among people, and often within the same person at different times.<sup>2</sup> Often, due to the difficulty of measuring feelings, customer satisfaction strategies are developed around clearly stated, logical customer opinions, and the emotional issues of a purchase are disregarded. This can be a costly mistake.

Customer satisfaction should not be viewed in a vacuum. For example, a customer may be satisfied with a product or service and therefore rate the product or service highly in a survey, and yet that same customer may buy another product or service. It is of little benefit to understand a customer's views about a product or service if the customer's views about competitors' product or service are not understood. The value customers place on one product compared to another may be a better indicator of customer loyalty. Customer loyalty can be sustained only by maintaining a favorable comparison when compared with competitors.<sup>3</sup> As mentioned before customer satisfaction is not a simple concept to understand or to measure.

## Who is the Customer?

There are two distinct types of customers—external and internal. An external customer can be defined in many ways, such as the one who uses the product or service, the one who purchases the product or service, or the one who influences the sale of the product or service. For instance, McDonald's determined the customer to be the child when they introduced their Happy Meals. The child never paid for the meals but the child influenced the sale. Oftentimes, parents purchase mobile phones and yet the teenage children use the mobile phones. The identity of the external customer is not always easy to determine.

<sup>2</sup> Jarrett Rosenberg, "Five Myths About Customer Satisfaction," *Quality Progress* (December 1996): 57–60.

<sup>3</sup> Robert Gardner, "What Do Customers Value," *Quality Progress* (November 2001): 41–48.

The leader's role is to process work through the internal customer-supplier chain by helping workers guarantee that the end product or service fully satisfies the end user. Rather than strive for personal objectives, each individual or group must identify and satisfy the internal customer(s) while fostering a team effort where all people help the organization. Each department must determine what activities are important to both external and internal customers and manage quality every step of the way. All quality management systems start with the basic need of ensuring that the external customer's requirements are adequately documented. Similarly, the organization must document explicitly what each internal customer expects. In addition, clear criteria must be provided for measuring success in meeting the expectations of both internal and external customers.

## Customer Perception of Quality

One of the basic concepts of the TQM philosophy is continuous process improvement. This concept implies that there is no acceptable quality level because the customer's needs, values, and expectations are constantly changing and becoming more demanding.

Before making a major purchase, some people check consumer magazines that rate product quality. During the period 1980 to 1988, the quality of the product and its performance ranked first, price was second, and service was third. During the period 1989 to 1992, product quality remained the most important factor, but service ranked above price in importance.

An American Society for Quality (ASQ) survey on end user perceptions of important factors that influenced purchases showed the following ranking.

1. Performance
2. Features
3. Service
4. Warranty
5. Price
6. Reputation

According to American Customer Satisfaction Index (ACSI), perceived quality is a measure of the customer's evaluation via recent consumption experience of the quality of a company's products or services. Quality is measured in terms of both customization, which is the degree to which a product or service meets the customer's individual needs, and reliability, which is the frequency with which things go wrong with the product or service.<sup>5</sup>

The factors of performance, features, service, and warranty are part of the product or service quality; therefore, it is evident that product quality and service are more important than price. Although this information is based on the retail customer, it appears, to some extent, to be true for the commercial customer also.

Thanks to the Internet, the world is now flooded with information available to the user in just a few seconds. Many customers would use online resources in order to compare various of product features and customer reviews before making a decision to buy. Thus buyers can influence decisions of potential buyers!

<sup>5</sup> <http://www.theacsi.org/about-acsi/the-science-of-customer-satisfaction>

# 4

## Employee Involvement

### Chapter Objectives

- Understanding motivation
- Empowering employees
- Assessing current levels through surveys
- Basics of Empowerment
- Understanding functioning and effectiveness of teams
- Encouraging employee participation through suggestion systems
- Understanding the concept of Gain sharing
- Brief overview of performance appraisals
- Overview of unions and employee involvement

### Introduction

Employee involvement is an approach to improving quality and productivity. Its use is credited for contributing to the success enjoyed by the Japanese in the world marketplace. Employee involvement is not a replacement for management nor is it the final word in quality improvement. It is a means to better meet the organization's goals for quality and productivity at all levels of an organization.

### Motivation

Knowledge of motivation helps us to understand the utilization of employee involvement to achieve process improvement.

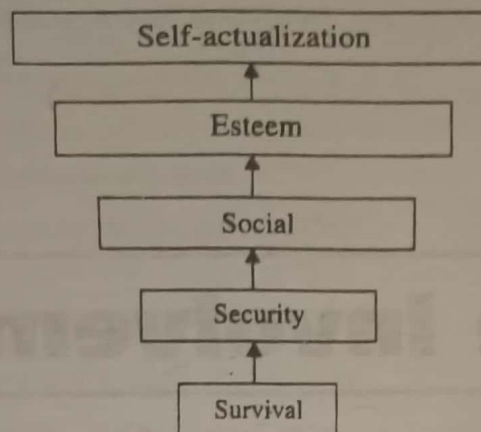


Figure 4-1 Maslow's Hierarchy of Needs

## Maslow's Hierarchy of Needs

One of the first and most popular motivational theories was developed by Abraham Maslow. He stated that motivation could best be explained in terms of a hierarchy of needs and that there were five levels. These levels are survival, security, social, esteem, and self-actualization. They are shown in Figure 4-1. Once a given level is satisfied, it can no longer motivate a person.

Relating these needs to motivation, we know that Level 1 (survival) means food, clothing, and shelter, which is usually provided by a job. In the workplace, Level 1 needs include proper lighting, heating/air conditioning, ventilation, phone system, data/voice access, and computer information system.<sup>1</sup> Level 2 (security) can mean a safe place to work and job security, which are very important to employees. When the organization demonstrates an interest in the personal well-being of employees, it is a motivating factor. A threat of losing one's job certainly does not enhance motivation. Level 2 is not limited to job security. It also includes having privacy on the job such as being able to lock one's office door or having lockable storage for personal items, as well as having a safe work environment that may include ergonomic adjustable furniture.<sup>2</sup>

Because we are social animals, Level 3 (social) relates to our need to belong. It has been said that cutting someone out of the group is devastating to that individual. Isolation is an effective punishment. Conversely, giving an individual the opportunity to be part of the group by feeling important and needed will motivate that person. If possible, employees should be provided with both formal social areas such as a cafeteria and conference rooms and informal areas such as water coolers and bulletin boards.<sup>3</sup> Being a member of a team is a good way to bring employees into the group. Level 4 (esteem) relates to pride and self-worth. Everyone, regardless of position or job assignment, wants to be recognized as a person of value to the organization. Where possible, employees should be given offices or personal spaces with aesthetics. Business cards, workspace size, and office protocols also provide employees with a certain level of self-esteem within an organization.<sup>4</sup> Seeking advice or

<sup>1</sup> Brenner, Pamela M., "Motivating Knowledge Workers: The Role of the Workplace," *Quality Progress* (January 1999): 33-37.

<sup>2</sup> Brenner.

<sup>3</sup> Brenner.

<sup>4</sup> Brenner.

There is nothing mystical about empowerment. People generally want to be more in charge of their jobs and careers. After all, they do that successfully in their personal lives every day. Most people appreciate and value the trust and responsibility inherent in an environment that supports empowered people and their actions. When people have the information, education, and skills required to perform in a changed environment, understand their boundaries of empowerment, and realize the necessity for change, their resistance to that change decreases greatly.<sup>17</sup>

A 1993 ASQ/Gallup survey showed that employees feel empowered to:

|                                      |     |
|--------------------------------------|-----|
| Stop work in progress. ....          | 83% |
| Intervene on customer's behalf ..... | 81% |
| Make exception to procedures .....   | 61% |
| Rework product or service .....      | 61% |
| Replace merchandise .....            | 37% |
| Refund money/authorize credit .....  | 26% |

## Teams

Employee involvement is optimized by the use of teams. Teams, however, are not a panacea for solving all quality and productivity problems, but in most instances, they are effective.

### Definition

A *team* is defined as a group of people working together to achieve common objectives or goals. *Teamwork* is the cumulative actions of the team during which each member of the team subordinates his individual interests and opinions to fulfill the objectives or goals of the group. The objective or goal is a need to accomplish something such as solve a problem, improve a process, design a refrigerator, plan a conference, audit a process, or please a customer. It needs to be clearly defined, have milestones set, have resources provided, and use a systematic approach. Members of the team will need to focus on how they relate to each other, listen to the suggestions of others, build on previous information, and use conflict creatively. They will need to set standards, maintain discipline, build team spirit, and motivate each other. Each member of the team has their own history of experience to help achieve the objective. They should have a need to see the task completed, but also the needs of companionship, fulfillment of personal growth, and self-respect.

### Why Teams Work

Teams work because many heads are more knowledgeable than one. Each member of the team has special abilities that can be used to solve problems. Many processes are so complex that one person cannot be knowledgeable concerning the entire process. Second, the whole is greater than the sum of its members. The interaction within the team produces results that exceed the contributions of each member. Third, team members develop a rapport with each other that allows them to do a better job. Finally, teams provide the vehicle for improved communication, thereby increasing the likelihood of a successful solution.

<sup>17</sup> Bob Mann, "Empowerment: An Enabling Process," *Quality Digest* (January 1994): 39-44.

# 5

## Continuous Process Improvement

### Chapter Objectives

- Brief overview: perfection versus continuous improvement
- Defining and mapping a process: learning from Juran Trilogy
- Overview of approaches for process improvement
- Choosing and integrating right improvement strategies
- Understanding the types of problems
- Understanding the concept of the PDCA cycle and problem solving method
- Studying the TQM approaches: Kaizen, Reengineering and Six Sigma

### Introduction

Quality-based organizations should strive to achieve perfection by continuously improving the business and production processes. Of course, perfection is impossible because the race is never over; however, we must continually strive for its attainment.

Improvement is made by

- Viewing all work as a process, whether it is associated with production or business activities.
- Making all processes effective, efficient, and adaptable.
- Anticipating changing customer needs.
- Controlling in-process performance using measures such as scrap reduction, cycle time, control charts, and so forth.
- Maintaining constructive dissatisfaction with the present level of performance.

- Eliminating waste and rework wherever it occurs.
- Investigating activities that do not add value to the product or service, with the aim of eliminating those activities.
- Eliminating nonconformities in all phases of everyone's work, even if the increment of improvement is small.
- Using benchmarking to improve competitive advantage.
- Innovating to achieve breakthroughs.
- Incorporating lessons learned into future activities.
- Using technical tools such as statistical process control (SPC), experimental design, benchmarking, quality function deployment (QFD), and so forth.

Continuous process improvement is designed to utilize the resources of the organization to achieve a quality-driven culture. Individuals must think, act, and speak quality. An organization attempts to reach a single-minded link between quality and work execution by educating its constituents to "continuously" analyze and improve their own work, the processes, and their work group.<sup>1</sup>

## Process

Process refers to business and production activities of an organization. Business processes such as purchasing, engineering, accounting, and marketing are areas where nonconformance can represent an opportunity for substantial improvement. Figure 5-1 shows a process model.

Inputs may be materials, money, information, data, etc. Outputs may be information, data, products, service, etc. The output of one process also can be the input to another process. Outputs usually

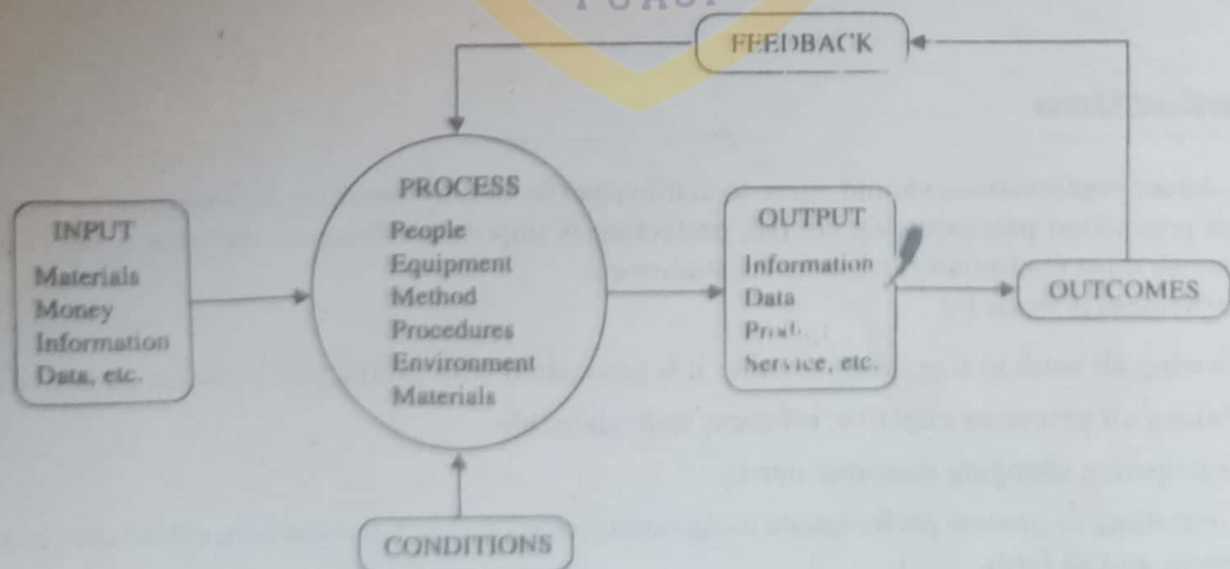


Figure 5-1 Input/Output Process Model

<sup>1</sup> Danny G. Langdon, "A New Language of Work," *Quality Digest* (October 1994): 44-48.

determine the root cause of the nonconformity and then take corrective action. Diagnosis can be difficult, because products and processes are quite complex. Standards cannot address all of the potential problems due to the interaction of individually-acceptable characteristics.

## Unstructured

Unstructured problems resemble compliance problems except that they are not specified by standards. The absence of standards may be due to system immaturity or to the need for flexibility in system performance. For example, an expert woodworker adjusts her activities to the grain and moisture content of the wood, and customer service workers adapt their behavior to individual customers. Identification of unstructured problems is usually brought about by negative customer feedback. The major challenges are to determine customer needs and to diagnose the causes of poor performance. Because of customer variability, it is difficult to determine why a product or service was unacceptable. Organizations need to treat each customer as an individual and maintain a database on acceptable and unacceptable behavior.

## Efficiency

Efficiency problems occur when the system is performing unacceptably from the viewpoint of its owners or operators. In other words, the end user is satisfied; however, the process is more costly than desired, or working conditions are not acceptable. Problem solving is directed towards reducing cost and providing safe working conditions. Identification of such problems occurs from benchmarking and operator suggestions.

## Process Design

Process-design problems involve the development of new processes and revision of existing processes. Many business and production processes have not been well designed or have become obsolete with advances in technology. Identification of problems is prompted by poor performance, the knowledge that we can do better (benchmarking), or the introduction of new products. It requires that user needs and relevant constraints be identified.

## Product Design

Product-design problems involve the development of new products and the improvement of existing products. A major focus is to prevent process and end user problems by relying on customer needs. Although design work can be initiated as a result of poor product performance, problem solving usually occurs as a natural part of a competitive environment. A major challenge is translating, in a timely manner, user needs and constraints into product attributes and specifications, usually using quality function deployment (QFD).

## The PDSA Cycle

The basic Plan-Do-Study-Act (PDSA) cycle was first developed by Shewhart and then modified by Deming.<sup>6</sup> It is an effective improvement technique. Figure 5-3 illustrates the cycle.

<sup>6</sup> Shewhart's cycle was called Plan-Do-Check-Act (PDCA).

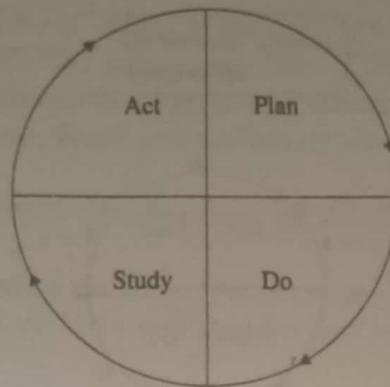


Figure 5-3 The PDCA Cycle

The four steps in the cycle are exactly as stated. First, *plan* carefully what is to be done. Next, carry out the plan (*do* it). Third, *study* the results—did the plan work as intended, or were the results different? Finally, *act* on the results by identifying what worked as planned and what didn't. Using the knowledge learned, develop an improved plan and repeat the cycle. The PDCA cycle is a simple adaptation of the more elaborate problem-solving method discussed in the next section.

## Problem-Solving Method

Process improvement achieves the greatest results when it operates within the framework of the problem-solving method. In the initial stages of a program, quick results are frequently obtained because the solutions are obvious or an individual has a brilliant idea. However, in the long term, a systematic approach will yield the greatest benefits.

The problem-solving method (also called the scientific method) has many variations depending, to some extent, on the use; however, they are all similar. There are seven phases, as shown in Figure 5-4. Also shown is the relationship to the PDCA cycle.

The phases are integrated because each phase is dependent upon the previous phase. Continuous process improvement is the objective, and these phases are the framework to achieve that objective.

### Phase 1: Identify the Opportunity

The objective of this phase is to identify and prioritize opportunities for improvement. It consists of three parts: identify the problem, form the team (if one is not in existence) and define the scope.

Problem identification answers the question, "What are the problems?" The answer leads to those problems that have the greatest potential for improvement and have the greatest need for solution. Problems can be identified from a variety of inputs, such as the following:

- Pareto analysis of repetitive external alarm signals, such as field failures, complaints, returns, and others (see Chapter 18).
- Pareto analysis of repetitive internal alarm signals (for example, scrap, rework, sorting, and the 100% test).
- Proposals from key insiders (managers, supervisors, professionals, and union stewards).

narrowed to five. Next, they prioritized the opportunities based on an estimate of the cost of poor quality. Upon collection of data, it was found that over half of the licensing applications received contained errors that doubled the processing time. Once the team identified this information, they developed an "opportunity statement" to "increase the percentage of error-free license applications from 40% to 70%." To analyze the root causes, the team flow-charted the licensing process and constructed a cause-and-effect diagram. Through a data collection effort, it was found that 50% of the errors were due to unqualified persons completing the license application, and that the license form and instructions were unclear. As a result, the team redesigned the application process so that the instructions were clearer and processing guidelines were understandable by the applicants. Before implementing the final solution, the form was pilot tested to a sample group. A little over a year, the team implemented the new form. Nine months after implementing their new solution, the percentage of applications received without errors increased from 40% to 80%, the backlog was reduced to 16 applications without staff increases, the speed of service improved 47%, and the cost of poor quality was reduced by 67%.<sup>11</sup>

Technology is also playing an important part in the management of continuous improvement. Many problem-solving approaches have been incorporated into software to help facilitate the continuous improvement process for organizations. According to a recent *Quality Progress* survey, 88% of the respondents indicated that their organization used software or similar technology to aid in the improvement process.<sup>12</sup>

## Kaizen

*Kaizen* is a Japanese word for the philosophy that defines management's role in continuously encouraging and implementing small improvements involving everyone. It is the process of continuous improvement in small increments that make the process more—efficient, effective, under control, and adaptable. Improvements are usually accomplished at little or no expense, without sophisticated techniques or expensive equipment. It focuses on simplification by breaking down complex processes into their sub-processes and then improving them.

The Kaizen improvement focuses on the use of:<sup>13</sup>

1. Value-added and non-value-added work activities.
2. *Muda*, which refers to the seven classes of waste—over-production, delay, transportation, processing, inventory, wasted motion, and defective parts.
3. Principles of motion study and the use of cell technology.
4. Principles of materials handling and use of one-piece flow.
5. Documentation of standard operating procedures.
6. The five S's for workplace organization, which are five Japanese words that mean proper arrangement (*seiko*), orderliness (*seiton*), personal cleanliness (*seiketso*), cleanup (*seiso*), and discipline (*shitsuke*).

<sup>11</sup> Fontaine, Daniel J. and Diane B. Robinette, "FCC Makes Dramatic Quality Improvements," *Quality Progress* (November 1994): 87–91.

<sup>12</sup> Brown, Paula, "Technology and Performance Improvement: Intellectual Partners?" *Quality Progress* (April 1998): 69–71.

<sup>13</sup> Glenn Gee, Phil McGrath, and Mahyar Izadi, "A Team Approach to Kaizen," *Journal of Industrial Technology* (Fall 1996): 45–48.

continuous improvement as referring to both incremental and “breakthrough” improvement. The Japanese have not only relied on kaizen but have developed policy management (*hoshin kanri*) and policy deployment (*hoshin tenkai*) in large part to produce the kind of large-scale breakthroughs that Hammer and Champy promote. Nor is this concept uniquely Japanese. Joseph Juran has had a long-standing emphasis on breakthrough efforts aimed at achieving unprecedented levels of performance.<sup>16</sup>

In 1997, EM Jorgensen Company applied reengineering using a five-phased problem solving approach that ultimately reduced operating costs by 12%. The focus of the project was to identify and eliminate non-value-added work and reduce corresponding costs while maintaining quality.<sup>17</sup>

## Six-Sigma\*

Six Sigma is one of the most popular approaches embraced by companies all over the world during the past two decades. Primary objective of Six Sigma is to reduce variation to achieve quality levels in products and processes to less than 3.4 defects per million opportunities (DPMO). Six Sigma approach is now deployed by many world class companies to make breakthrough improvements that directly lead to improved financials. It has now become a model for business excellence. In India, Six Sigma approach is deployed by companies such as Cummins India, SKF, TCS, Bharat Electronics Limited (BEL), Bharat Heavy Electricals, Tata Motors, Reliance, Capgemini, John Deere, Alfa Laval, Asian Paints, etc.

The Six Sigma approach was first introduced and developed at Motorola in the late 1980s. Motorola learned about quality the hard way as it was being consistently beaten in the competitive marketplace.<sup>18</sup> A Japanese firm took over a Motorola factory, which made television sets in the United States. The Japanese management improved the quality performance and reduced the defect level of the television sets to 1/20th of the number of defects they had produced with Motorola management.

The senior management of Motorola realized the urgent need to improve their quality performance and reputation. In 1986, Bill Smith, a senior engineer and scientist within Motorola's Communications Division, introduced the concept of Six Sigma in response to increasing complaints from the field sales force about warranty claims. Smith approached Bob Galvin, the CEO at the time, and explained his analysis and ideas of Six Sigma approach. Bob recognized the power of the approach and decided to adapt it for Motorola.<sup>19</sup> Today, Bill Smith is known as the ‘Father of Six Sigma’<sup>20</sup>. Born in Brooklyn, New York, Smith graduated from the U.S. Naval Academy in 1952 and studied at the University of Minnesota School of Management (now known as the Carlson School of Management).

While Six Sigma was developed at Motorola, the concepts, approach and tools have been evolved gradually through knowledge, experience and wisdom of many quality experts and statisticians. Before Six Sigma was ‘born’, industry had tried and tested Total Quality Management (TQM). Many companies had significant success with TQM. The roots of Six Sigma can therefore be traced back

\* This section is adapted from ‘Six Sigma for Business Excellence’ by Hernant Urdhwarshie, published by Pearson Education.

<sup>16</sup> Robert E. Cole, “Reengineering the Corporation: A Review Essay,” *Quality Management Journal* (July 1994): 77–85.

<sup>17</sup> Dagestino, Kathryn, Wendy L. Moore, and Jorn Teutloff, “Hustle, That’s All,” *Quality Progress* (September 2000): 73–79.

<sup>18</sup> Thomas Pyzdek, ‘The Six Sigma Handbook: A Complete Guide for Green Belts, Black Belts and Managers at All Levels’ (2003).

<sup>19</sup> Tom McCarty, ‘Six Sigma® at Motorola,’ European CEO, September–October 2004. Leadership Edition (2004). [www.motorola.com/mot/doc/1/1736\\_MotDoc.pdf](http://www.motorola.com/mot/doc/1/1736_MotDoc.pdf). Accessed on 26 Nov 2009

<sup>20</sup> Gail Chadwick, ‘Remembering Bill Smith, Father of Six Sigma,’ (2000), published by [www.isixsigma.com](http://www.isixsigma.com)

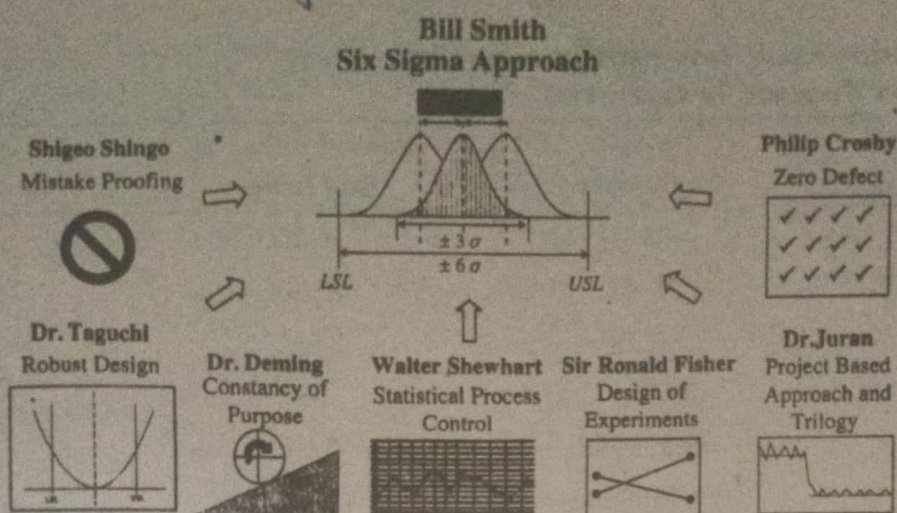


Figure 5-5 Six Sigma Approaches

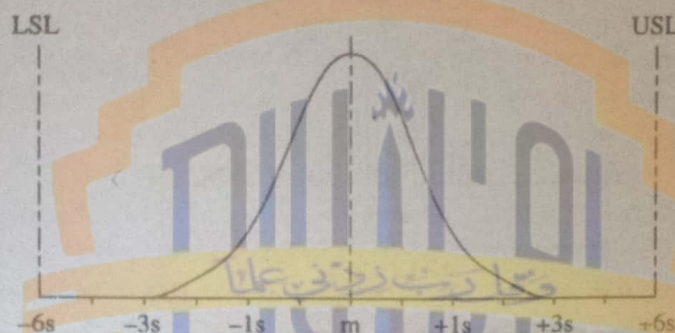


Figure 5-6 Nonconformance Rate When Process is Centered

to the philosophies of many quality management gurus. These philosophies were part of the TQM approach as well. While it is impossible to include all contributions the most prominent ones are included in illustration shown in Figure 5.5.

## Statistical Aspects

According to James Harrington, "Six sigma was simply a TQM process that uses process capability analysis (see Chapter 18) as a way of measuring progress."<sup>21</sup> Sigma,  $\sigma$ , is the Greek symbol for the statistical measurement of dispersion called standard deviation. It is the best measurement of process variability, because the smaller the deviation value, the less variability in the process. Figure 5-6 shows a process that is normally distributed and centered with the upper and lower specification limits (USL and LSL) established at  $\pm 6\sigma$ . For this situation, 99.999998% of the product or service will be between specifications, and the nonconformance rate will be 0.002 parts per million, or 2.0 per billion. The situation diagrammed represents a process capability index ( $C_p$ ) of 2.0. A  $C_p$  of 1.33 has been a defacto standard. Table 5-4 shows the percent between specifications, the nonconformance rate, and process capability for different specification limit locations.

<sup>21</sup> H. James Harrington, "Six Sigma's Long-Term Impact," *Quality Digest* (June 2001): 16.

# 8

## Benchmarking

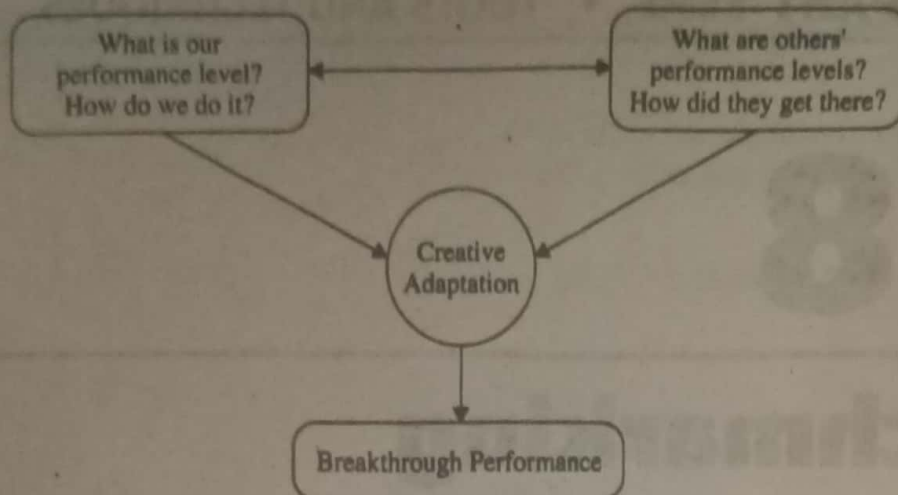
### Chapter Objectives

- Defining benchmarking and understanding its need
- Understanding the process of benchmarking
- Identifying the critical factors to success
- Understanding the current performance
- Planning the benchmarking exercise
- Studying and analyzing the results
- Understanding limitations and pitfalls

### Introduction

Benchmarking is a systematic method by which organizations can measure themselves against the best industry practices. It promotes superior performance by providing an organized framework through which organizations learn how the "best in class" do things, understand how these best practices differ from their own, and implement change to close the gap. The essence of benchmarking is the process of borrowing ideas and adapting them to gain competitive advantage. It is a tool for continuous improvement.

Benchmarking is an increasingly popular tool. It is used extensively by both manufacturing and service organizations, including Xerox, AT&T, Motorola, Ford, and Toyota. Benchmarking is a common element of quality standards, such as the Chrysler, Ford, and General Motors Quality System Requirements. These standards stipulate that quality goals and objectives be based on competitive products and benchmarking, both inside and outside the automotive industry. The Malcolm Baldrige National Quality Award similarly requires that applicants benchmark external organizations.



**Figure 8-1 Benchmarking Concept**

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## Benchmarking Defined

Benchmarking is the systematic search for best practices, innovative ideas, and highly effective operating procedures. Benchmarking considers the experience of others and uses it. Indeed, it is the common-sense proposition to learn from others what they do right and then imitate it to avoid reinventing the wheel. Benchmarking is not new and indeed has been around for a long time. In fact, in the 1800s, Francis Lowell, a New England colonist, studied British textile mills and imported many ideas along with improvements he made for the burgeoning American textile mills.

As shown in Figure 8-1, benchmarking measures performance against that of best-in-class organizations, determines how the best in class achieve those performance levels, and uses the information as the basis for adaptive creativity and breakthrough performance.<sup>1</sup>

Implicit in the definition of benchmarking are two key elements. First, measuring performance requires some sort of units of measure. These are called metrics and are usually expressed numerically. The numbers achieved by the best-in-class benchmark are the target. An organization seeking improvement then plots its own performance against the target. Second, benchmarking requires that managers understand why their performance differs. Benchmarkers must develop a thorough and in-depth knowledge of both their own processes and the processes of the best-in-class organization. An understanding of the differences allows managers to organize their improvement efforts to meet the goal. Benchmarking is about setting goals and objectives and about meeting them by improving processes.

## Reasons to Benchmark

Benchmarking is a tool to achieve business and competitive objectives. It is powerful and extremely effective when used for the right reasons and aligned with organization strategy. It is not a panacea that can replace all other quality efforts or management processes. Organizations must still decide which

<sup>1</sup> Christopher E. Bogan and Michael J. English, "Benchmarking for Best Practices: Winning Through Innovative Adaptations," *Quality Digest* (August 1994): 52-62.

markets to serve and determine the strengths that will enable them to gain competitive advantage. Benchmarking is one tool to help organizations develop those strengths and reduce weaknesses.

By definition, benchmarking requires an external orientation, which is critical in a world where the competitor can easily be on the other side of the globe. An external outlook greatly reduces the chance of being caught unaware by competition. Benchmarking can notify the organization if it has fallen behind the competition or failed to take advantage of important operating improvements developed elsewhere. In short, benchmarking can inspire managers (and organizations) to compete.

In contrast to the traditional method of extrapolating next year's goal from last year's performance, benchmarking allows goals to be set objectively, based on external information. When personnel are aware of the external information, they are usually much more motivated to attain the goals and objectives. Also, it is hard to argue that an objective is impossible when it can be shown that another organization has already achieved it.

Benchmarking is time and cost efficient because the process involves imitation and adaptation rather than pure invention. Benchmarking partners provide a working model of an improved process, which reduces some of the planning, testing, and prototyping effort. As the old saying goes, Why reinvent the wheel?

The primary weakness of benchmarking, however, is the fact that best-in-class performance is a moving target. For example, new technology can create quantum leap performance improvements, such as the use of electronic data interchange (EDI). Automobile makers no longer use paper to purchase parts from suppliers. A computer tracks inventory and transmits orders directly to a supplier's computers. The supplier delivers the goods, and payment is electronically transmitted to the supplier's bank. Big Bazar in India runs SAP ERP system for their inventory management. These applications of EDI save tens of thousands of worker hours and whole forests of trees, as well as helping to meet customer requirements.

For functions that are critical to the business mission, organizations must continue to innovate as well as imitate. Benchmarking enhances innovation by requiring organizations to constantly scan the external environment and to use the information obtained to improve the process. Potentially useful technological breakthroughs can be located and adopted early.

## Process

Organizations that benchmark, adapt the process to best fit their own needs and culture. Although the number of steps in the process may vary from organization to organization, the following six steps contain the core techniques.

1. Decide what to benchmark.
2. Understand current performance.
3. Plan.
4. Study others.
5. Learn from the data.
6. Use the findings.

Table 8-1 illustrates how AT&T and Xerox have adapted benchmarking to their own needs. AT&T, in its first six steps, explicitly incorporates training and makes sure that personnel using benchmarking results to improve their processes buy into the program. The assumption is that if the process owners are not committed, they will ignore the results and the effort will have been wasted. Steps 7 through 12 represent the core benchmarking process.

# 10

## Quality Management Systems

### Chapter Objectives

- Understanding the ISO 9000 series of standards, namely, ISO 9000, ISO 9001 and ISO 9004
- Brief overview of sector specific standards, namely, AS 9100, ISO/TC 16946 and TL 9000
- Understanding the ISO 9001 requirements
- Overview of ISO 9001 implementation
- Detailed analysis of registration requirements

### Introduction

The International Organization for Standardization (ISO) was founded in 1946 in Geneva, Switzerland, where it is still based. Its mandate is to promote the development of international standards to facilitate the exchange of goods and services worldwide. ISO is composed of more than 90 member countries. The United States representative is the American National Standards Institute (ANSI).

The ISO Technical Committee (TC) 176 developed a series of international standards for quality systems, which were first published in 1987. The standards (ISO 9000, 9001, and 9004) were intended to be advisory and were developed for use in two-party contractual situations and internal auditing. However, with their adoption by the European Community (EC) and a worldwide emphasis on quality and economic competitiveness, the standards have become universally accepted.

The fourth edition of ISO 9001 was released in the year 2008 and it replaces the third edition (ISO 9001:2000), which had been amended to clarify the points in the text and also to enhance the compatibility with ISO 14001:2004.

Most countries have adopted the ISO 9000 series as their national standards. Likewise, thousands of organizations throughout the world have quality systems registered to the standard. In the United States, the national standards are published by the American National Institute/American Society for Quality (ANSI/ASQ) as the ANSI/ASQ Q9000 series. Government bodies throughout the world, including the United States, are also using the standards. U.S. government agencies using the series include the Department of Defense (DOD) and the Food and Drug Administration (FDA).

In India, Bureau of Indian Standards (BIS) adopts ISO certification standards under the dual numbering scheme.

In a two-party system, the supplier of a product or service would develop a quality system that conformed to the standards. The customers would then audit the system for acceptability. This two-party system results in both the supplier and customer having to participate in multiple audits, which can be extremely costly. This practice is replaced by a third-party registration system.

A quality system registration involves the assessment and periodic surveillance audit of the adequacy of a supplier's quality system by a third party, who is a registrar. When a system conforms to the registrar's interpretation of the standard, the registrar issues a certificate of registration to the supplier. This registration ensures customers or potential customers that a supplier has a quality system in place and it is being monitored.

## Benefits of ISO Registration

There are various reasons for implementing a quality system that conforms to an ISO standard. The primary reason is that customers or marketing are suggesting or demanding compliance to a quality system. Other reasons are needed improvement in processes or systems and a desire for global deployment of products and services.<sup>1</sup> As more and more organizations become registered, they are requiring their subcontractors or suppliers to be registered, creating a snowball effect. Consequently, in order to maintain or increase market share, many organizations are finding they must be in conformance with an ISO standard. Internal benefits that can be received from developing and implementing a well-documented quality system can far outweigh the external pressures.

A study of 100 Italian manufacturing firms was undertaken to determine if there was any improvement in performance after registration. Significant improvement was noted in:

- Internal quality as measured by the percent of scrap, rework, and nonconformities at final inspection.
- Production reliability as measured by the number of breakdowns per month, percent of time dedicated to emergencies, and percent of downtime per shift.
- External quality as measured by product accepted by customers without inspection, claims of nonconforming product, and returned product.
- Time performance as measured by time to market, on-time delivery, and throughput time.
- Cost of poor quality as measured by external nonconformities, scrap, and rework.

<sup>1</sup> F. C. Weston, Jr., "What Do Managers Really Think of the ISO 9000 Registration Process?" *Quality Progress* (October 1995): 67-73.

On the negative side, prevention and appraisal costs increased.<sup>2</sup>

Additional examples of benefits after registration are:

The American Institute of Certified Public Accountants (AICPA) now has a quality system that works, and there was a 4% improvement in gross margins, which was the largest improvement in their history.<sup>3</sup>

## ISO 9000 Series of Standards

The ISO 9000 Series of Standards is generic in scope. By design, the series can be tailored to fit any organization's needs, whether it is large or small, a manufacturer or a service organization. It can be applied to construction, engineering, health care, legal, and other professional services as well as the manufacturing of anything from nuts and bolts to spacecraft. Its purpose is to unify quality terms and definitions used by industrialized nations and use those terms to demonstrate a supplier's capability of controlling its processes. In very simplified terms, the standards require an organization to say what it is doing to ensure quality, then do what it says, and, finally, document or prove that it has done what it said.

The three standards of the series are described briefly in the following paragraphs:

*ISO 9000:2005—Quality Management Systems (QMS)—fundamentals and vocabulary* discusses the fundamental concepts related to the QMS and provides the terminology used in the other two standards.

*ISO 9001:2008—Quality Management Systems (QMS)—requirements* is the standard used for registration by demonstrating conformity of the QMS to customers, regulatory, and the organization's own requirements.

*ISO 9004:2000—Quality Management Systems (QMS)—guidelines for performance improvement* provides guidelines that an organization can use to establish a QMS focused on improving performance.

## Sector-specific Standards

The ISO 9000 system is designed as a simple system that could be used by any industry. Other systems have been developed that are specific to a particular industry such as automotive or aerospace. These systems use the ISO 9001 as the basic framework and modify it to their needs. There are currently three other quality systems: AS9100, ISO/TS 16949, and TL 9000.

One of the problems with sector-specific standards is the need for suppliers with customers in different industries to set up quality systems to meet each sector's requirements. For example, a packaging supplier that services the aerospace, automobile, and telecommunications industries would need to set up its system to accommodate not only ISO 9001 but three other standards. In addition, the Registration Accreditation Board (RAB) points out that sector-specific standards have created a need for specialized auditors and training courses. On the positive side, the standardization of requirements beyond ISO 9001 makes compliance by key suppliers and implementation by major customers much easier.<sup>4</sup>

<sup>2</sup> Pietro Romano, "ISO 9000: What Is Its Impact On Performance?" *Quality Management Journal* (Vol. 7, No. 3, 2000): 38–55.

<sup>3</sup> Norman Ho, "ISO 9000: No Longer a Stranger to Service," *Quality Digest* (June 1999): 33–36.

<sup>4</sup> Leslie Norris, "The Pros and Cons of Sector-Specific Standards," *Quality Progress* (April 1999): 92–3.

Growth in the use of management systems generally has increased the need to ensure that laboratories, which form part of larger organizations or offer other services can operate to a quality management system that is seen as compliant with ISO 9001 as well as with this international standard.

Conformity to the quality management system within which the laboratory operates to the requirements of ISO 9001 does not of itself demonstrate the competence of the laboratory to produce technically valid data and results. Nor does demonstrated conformity to this international standard imply conformity of the quality management system within which the laboratory operates to all the requirements of ISO 9001.

Testing and calibration laboratories that comply with this international standard will therefore also operate in accordance with ISO 9001. Clause 4 of this standard specifies the requirements for sound management. Clause 5 specifies the requirements for technical competence for the type of tests and/or calibrations the laboratory has undertaken. Accreditation bodies that recognize the competence of testing and calibration laboratories are expected to use this international standard as the basis for their accreditation.

The acceptance of testing and calibration results between countries is expected to be facilitated if laboratories comply with this international standard and if they obtain accreditation from bodies which have entered into mutual recognition agreements with equivalent bodies in other countries using this international standard.

In India, accreditation of laboratories as per ISO/IEC 17025 is done by NABL (National Accreditation Board for Testing and Calibration Laboratories). NABL is an autonomous body under the aegis of Department of Science and Technology, Government of India.

NABL is a signatory to Asia Pacific Laboratory Accreditation Cooperation (APLAC), Mutual Recognition Agreement (MRA) and International Laboratory Accreditation Cooperation (ILAC) Arrangement. Thus the test results and measurement data produced by Indian accredited laboratories are acceptable amongst 64 economies worldwide.

## ISO 15189: 2012

This standard specifies requirements for quality and competence in medical laboratories. ISO 15189 can be used by medical laboratories in developing their quality management system and assessing their own competence. It can also be used for conforming or recognising the laboratory by their customers, regulatory authorities and accreditation bodies. In India, NABL acts as an accreditation body as per ISO 15189.

## ISO 9001 Requirements<sup>9</sup>

The standard has eight clauses: Scope, Normative References, Definitions, Quality Management Systems, Management Responsibility, Resource Management, Product and/or Service Realization, and Measurement, Analysis, and Improvement. The first three clauses are for information while the last five are requirements that an organization must meet. The numbering system used in the standard is followed in this section.

The application of a system of processes within an organization, together with their identification and interactions and the managing of these processes, is referred to as the process approach. This approach emphasizes the importance of:

<sup>9</sup> Adapted, with permission, from *Quality Management Systems—Requirements*, ANSI/ISO/ASQ Q9001: 2000, © 2000 ASQ Quality Press. (Milwaukee, WI: ASQ 2000).

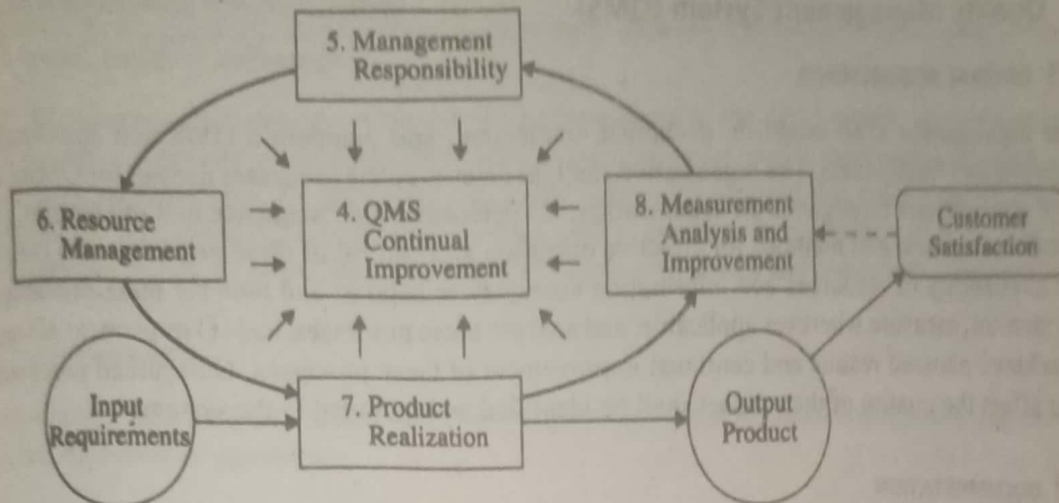


Figure 10-2 Model of a Process-Based Quality Management System

- Understanding and fulfilling the requirements.
- The need to consider processes in terms of value added.
- Obtaining results of process performance and effectiveness.
- Continual improvement of processes based on objective measure.

For the five required clauses, the system is shown in Figure 10-2.

## 1. Scope

The purpose of the standard is for the organization to demonstrate its ability to provide a product<sup>10</sup> that meets customer and regulatory requirements and to enhance customer satisfaction. This purpose is accomplished by evaluating and continually improving the system, rather than the product. The requirements of the standard are intended to be applicable to all types and sizes of organizations. Requirements in Clause 7, Product Realization, that are not appropriate to the organization, can be excluded. Such exclusions do not affect the organization's ability or responsibility to provide products that meets customer, applicable statutory and regulatory requirements.

## 2. Normative Reference

ISO 9000:2005 *Quality Management Systems—Fundamentals and vocabulary* are a normative reference that provides applicable concepts and definitions.

## 3. Terms and Definitions

For the purposes of this standard, the terms and definitions given in ISO 9000:2000 apply. Throughout the text of this standard, the term "Product" also means "Service."

<sup>10</sup> The standard defines product as including both product and service.

auditing, environmental labeling, environmental performance evaluation, life-cycle assessment, and terms and definitions. In addition, a working group on environmental aspects in product standards was formed. Each country has a member body of ISO, which for the United States is the American National Standards Institute (ANSI) and, for India, it is Bureau of Indian Standards (BIS).

Experience with ISO 9000 made it easier to develop ISO 14000, and a subsequent revision of ISO 9000 capitalized on many of the improvements in ISO 14000. The EMS is part of a comprehensive management system that addresses how the overall business activities, including its products and services, impact the environment. The EMS maximizes company participation in environmental performance now and in the future. Processes for obtaining registration closely resemble those involved with ISO 9000 and related quality standards. The differences lie mainly in what the standard requires rather than the registration process.<sup>1</sup>

IS/ISO 14001 was originally published in 1997 in India under the dual numbering scheme of BIS adopting first edition of ISO 14001. With the publication of second edition of ISO 14001 in 2004, the revision of IS/ISO 14001 was taken up by BIS.

## ISO 14000 Series Standards

The series is divided into two separate areas—the organization evaluation standards and the product evaluation standards. In addition, ISO 14050 covers terms and definitions that are common to both areas. Most recent version of this standard was issued recently in 2009.

### Organizational Evaluation Standards

These standards are shown in Figure 11-1 and consist of three categories: Environmental Management System (EMS), Environmental Auditing (EA), and Environmental Performance Evaluation (EPE).

ISO 14001, entitled, “Environmental Management Systems—Specifications with Guidance for Use,” gives the elements that organizations are required to conform to if they seek registration. This standard is the heart of the standards and will be discussed in greater detail later in the chapter. The standards given below support the EMS.

ISO 14004, entitled, “Environmental Management Systems—Guidelines on Principles, Systems, and Supporting Techniques,” provides supplementary material. It is for information only and is not to be used for registration.

ISO 14010, ISO 14011 and ISO 14012 standards are now superseded by ISO 19011:2002 (Guidelines for quality and environmental management system auditing), which covers a range of audit related functions. It is intended that by using this new standard, organizations can save time and money by:

- Avoiding confusion over objectives of environmental or quality audit programs.
- Establishing agreement of goals between individual audit programs.
- Reducing duplication of efforts.
- Ensuring the use of best audit formats.
- Evaluating competence of audit team members against appropriate criteria.

<sup>1</sup> Joe Lissenden, “ISO 9000 Eases ISO 14001 Registration,” *Quality Digest* (May 1999): 41–44.

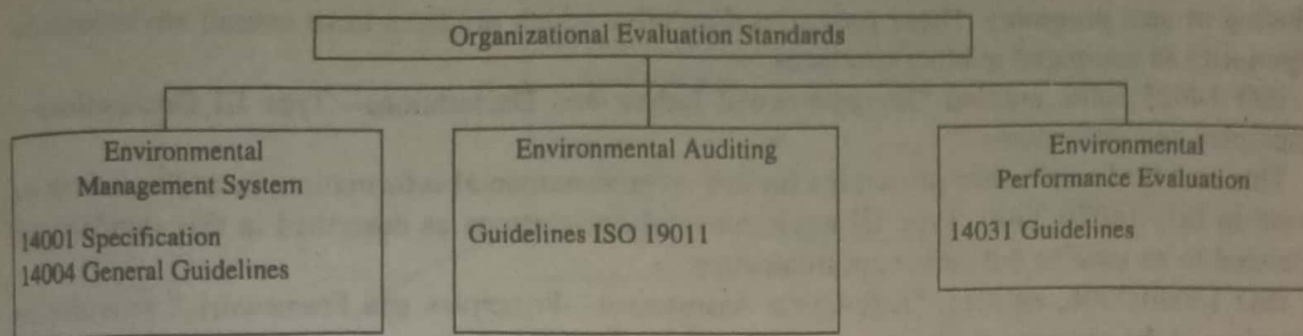


Figure 11-1 Organizational Evaluation Standards

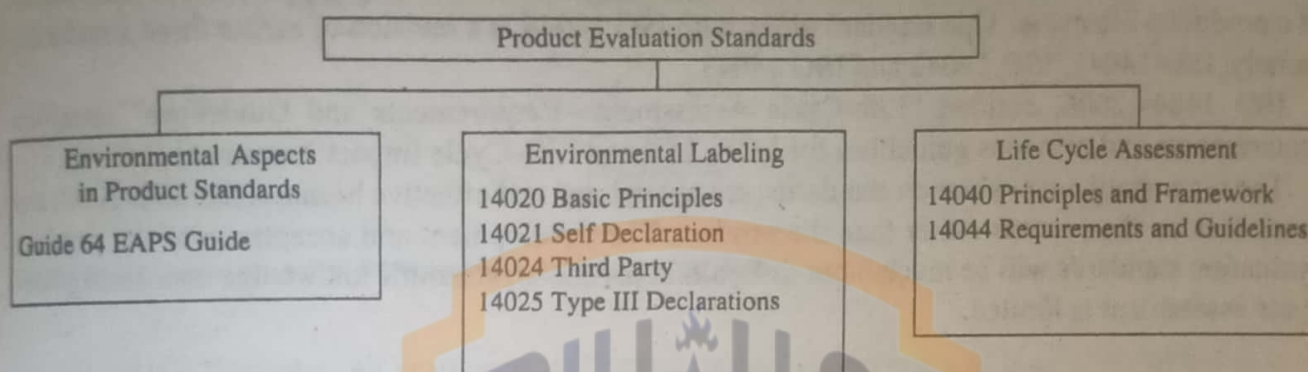


Figure 11-2 Product Evaluation Standards

ISO 14031, entitled, "Guidelines on Environmental Performance Evaluation," presents information on recording information to track performance. It helps the organization meet the requirements of ISO 14001, Section 4.5.1, Monitoring and Measuring.

## Product Evaluation Standards

These standards are under development and are shown in Figure 11-2. They consist of three categories: Environmental Aspects in Product Standards (EAPS), Environmental Labeling (EL), and Life-Cycle Assessment (LCA).

ISO Guide 64:2008 entitled, "Environmental Aspects in Product Standards," is designed to help writers develop product standards. Writers should carefully consider the environmental consequences when developing criteria, elements, and characteristics that go into the standard.

ISO 14020:2000, entitled, "Environmental Labels and Declarations—General Principles for All Environmental Labeling," provides guidance on the goals and principles that should be used in development and use of environmental labels and declarations. Product improvement is a desirable benefit but not the objective of the standard. This standard is not intended for use as a specification for certification and registration purposes.

ISO 14021:1999 entitled, "Environmental Labeling—Self-Declaration of Environmental Claims: Terms and Definitions," applies to organizations that are declaring that their product has an environmental attribute such as being recyclable or energy efficient. The standard ensures that this type of labeling is accurate, verifiable, and not deceptive.

ISO 14024, entitled, "Environmental Labeling—Practitioner Programs: Guiding Principles, Practices, and Certification Procedures for Multiple Criteria Programs," establishes criteria for third-party

labeling or seal programs. These programs determine which products have overall environmental superiority as compared to other products.

ISO 14025:2006, entitled "Environmental Labels and Declarations—Type III Declarations—Principles and Procedures."

This standard establishes principles for use of environmental information, in addition to those given in ISO 14020:2000. Type III environmental declarations as described in this standard are intended to be used in business communication.

ISO 14040:2006, entitled, "Life-Cycle Assessment—Principles and Framework," provides an overview of the practice, applications, and limitations of LCA. Life-cycle assessment and life-cycle invest (LCI) attempt to determine the long-range environmental effect of a product. This assessment is an enormously difficult task because of the unforeseen and frequently controversial nature of a product's life cycle. This standard along with ISO 14044 is a revision of earlier three standards, namely, ISO 14041, ISO 14042 and ISO 14043.

ISO 14044:2006, entitled "Life-Cycle Assessment—Requirements and Guidelines" specifies requirements and provides guidelines for LCA, LCI and Life-Cycle Impact Assessment (LCIA).

The organizational evaluation standards, are operational and effective because, like ISO 9000, the focus was on the process rather than the product. The development and acceptance of the product evaluation standards will be much more difficult. In particular, scientific knowledge concerning life-cycle assessment is limited.

## Concepts of ISO 14001

This standard provides organizations with the elements for an environmental management system (EMS), which can be integrated into other management systems to help achieve environmental and economic goals. It describes the requirements for registration and/or self-declaration of the organization's EMS. Demonstration of successful implementation of the system can be used to assure other parties that an appropriate EMS is in place. It was written to be applicable to all types and sizes of organizations and to accommodate diverse geographical, cultural, and social conditions. As previously mentioned, the requirements are based on the process and not on the product. It does, however, require commitment to the organization's EMS policy, applicable regulations, and continual improvement.

The basic approach to EMS is shown in Figure 11-3. It begins with the environmental policy, which is followed by planning, implementation and operation, checking and corrective action, and management review. The approach follows the PDSA cycle. There is a logical sequence of events to achieve continual improvement. Many of the requirements may be developed concurrently or revisited at any time. The overall aim is to support environmental protection and prevention of pollution in balance with socioeconomic needs.

The standard is not intended to create nontariff barriers or to change an organization's legal obligations. In addition, it does not include aspects of occupational health and safety management, although an organization may include these aspects in the documentation.

In order to understand the requirements, a few definitions are necessary. *Environment* is defined as the global surroundings in which an organization operates and includes air, water, land, natural resources, flora, fauna, humans, and their interaction. *Environmental aspect* is defined as an element of an organization's activities, products, or services that can interact with the environment. Examples

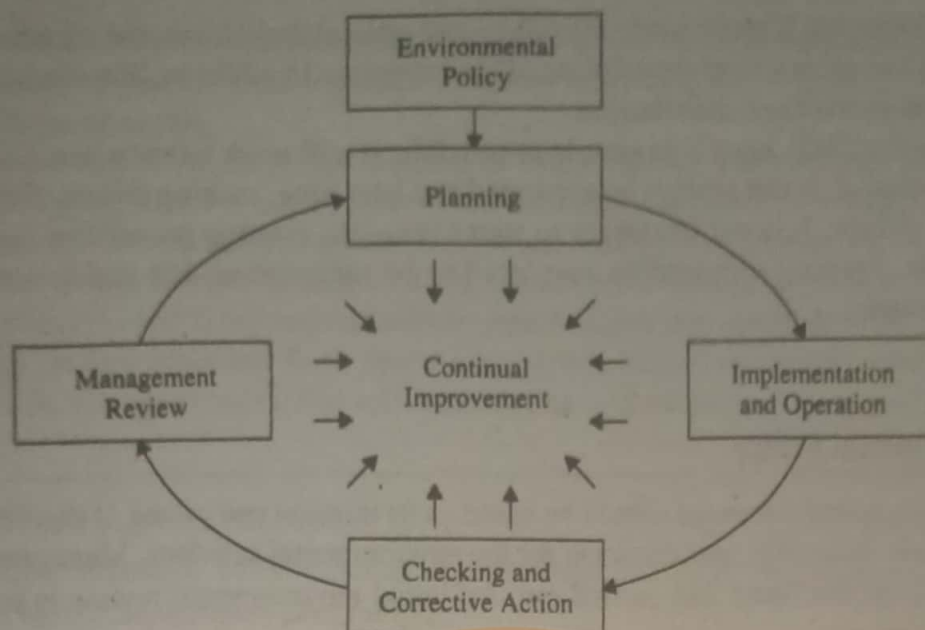


Figure 11-3 Environmental Management System Model

are wastewater discharge, air emissions, and energy usage. *Environmental impact* is defined as any change, whether adverse or beneficial, wholly or partially resulting from an organization's environmental aspects. Examples are impacts on habitat, water supply, and soil erosion. *Environmental objective* is an overall environmental goal, arising from the policy statement, that an organization sets for itself and which is quantified when practical. They define how the policy will be achieved. For example, an objective could be to control the temperature of the wastewater effluent. *Environmental target* is a detailed performance requirement that arises from the environmental objectives. For example the wastewater temperature should be controlled between 10° C and 14° C.

There are four sections to the standard—scope, normative references, definitions, and EMS requirements—and an informative annex. The EMS requirements are given in Section 4 of the standard and are covered next.

## Requirements of ISO 14001<sup>2</sup>

The standard is divided into six parts or clauses and has a total of 18 requirements. The numbering system used is identical to the standard OHSAS 18001.

### 4.1 General Requirements

The organization shall establish, maintain and continually improve an environmental management system that includes policy, planning, implementation and operation, checking and corrective action, and management review. These requirements are given in the rest of the standard. The organization shall define and document the scope of its environment management system.

<sup>2</sup> This section adapted from ANSI/ISO 14001-2004, *Environmental Management Systems—Specification with Guidance for Use*, Department of Environmental Quality, Michigan Government.

Evaluation of the continued suitability of the policy in light of changing legislation; changing expectations; changing requirements of interested parties; changes in activities, products, and services; new technology; lessons learned; market preferences and expectations; and effectiveness of reporting and communication.

Reviews should make use of information from audit reviews, performance information, changing circumstances, and the commitment to continuous improvement.

## Benefits of EMS

EMS benefits can be categorized as global and organizational.

### Global<sup>4</sup>

There are three global benefits: (1) facilitate trade and remove trade barriers, (2) improve environmental performance of planet earth, and (3) build consensus that there is a need for environmental management and a common terminology for EMS.

The proliferation of national and regional standards has led to confusion and to trade barriers. This international standard will serve to unify countries in their approach to labeling, environmental management, and life-cycle assessment. This approach also will help to remove trade barriers and facilitate trade. The drafters of the standard have been careful to avoid creating a document that will hamper trade. ISO 14000 provides the framework whereby its successful promotion within countries can lead to progress that will reassure the worldwide community. As the EMS is implemented worldwide, it will increasingly satisfy concerns for environmental protection in trade discussions and agreements.

Although the standard does not specify performance, it is reasonable to expect that ISO 14000 will play a significant role in environmental improvement worldwide. Because of the success of ISO 9000 in improving quality, we can reasonably expect that a significant improvement will occur in the environment. Based on data collected during registration audits, many ISO 14001 EMS lead auditors are convinced that an EMS does improve regulatory compliance.<sup>5</sup>

As the environmental standards are developed and implemented worldwide, it will build worldwide consensus that there is a need to use ISO 14000. In addition, a common terminology allows people in different countries to speak to each other, thereby sharing improvement ideas, prevention information, and system problems.

### Organizational

According to ISO 14004, an organization benefits from an EMS in a number of ways as given below:

- Assuring customers of a commitment to environmental management.
- Meeting customer requirements, the primary reason for organizations to become certified.
- Maintaining a good public/community relations image.
- Satisfying investor criteria and improving access to capital.

<sup>4</sup> This section adapted from Joseph Cascio, Gayle Woodside, and Philip Mitchell, *ISO 14000 Guide* (New York: McGraw-Hill, 1996).

<sup>5</sup> Marilyn Block, "ISO 14001 and Regulatory Compliance," *Quality Progress* (February 2001): 84-85.